

PORTO SEGURO, Brazil

WMO: 834600

Lat: 16.439S

Lon: 39.081W

Elev: 168

StdP: 14.61

Time Zone: -3.00 (W03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	62.3	64.0	58.7	74.1	68.0	60.4	78.8	68.3	16.5	75.0	15.1	75.9	5.5	270	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.2	87.9	77.0	86.4	76.5	85.9	76.3	78.6	84.5	78.1	83.8	77.5	83.2	15.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	141.7	81.6	75.6	135.0	80.5	75.5	134.3	80.4	42.2	85.0	41.5	83.8	41.0	83.2	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	18.3	16.4	60.4	90.9	2.5	1.5	58.6	91.9	57.1	92.8	55.7	93.6	53.9	94.7		
			59.1	80.2	2.8	1.6	57.1	81.4	55.5	82.3	53.9	83.2	51.9	84.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.7	78.4	78.8	78.9	77.4	75.0	73.3	71.9	71.8	73.2	75.3	76.6	78.1
	DBStd	3.45	1.83	1.89	2.09	2.36	2.36	2.51	2.06	2.32	2.57	2.57	2.77	2.13
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDD50	9386	880	807	896	823	775	700	680	676	697	783	797	872
	CDD65	3912	415	387	431	373	310	250	215	212	247	318	347	407
	CDH74	26024	3529	3577	3735	2785	1667	1002	686	719	1122	1823	2217	3162
	CDH80	5033	811	922	985	607	214	73	18	21	78	218	389	697
(14) Wind	WSAvg	7.4	8.2	7.7	7.3	6.3	5.8	5.8	6.2	6.9	8.3	8.9	8.7	8.9
(15) Precipitation	PrecAvg	69.6	3.8	4.0	4.9	7.8	6.6	5.4	6.8	4.1	5.8	6.7	5.5	5.6
	PrecMax	90.8	11.4	9.8	9.3	12.7	14.6	11.5	19.2	8.8	14.6	13.6	17.2	12.6
	PrecMin	39.6	0.4	0.4	0.4	2.5	0.4	1.7	1.9	1.6	2.0	1.6	1.0	0.5
	PrecStd	14.6	3.1	2.8	2.2	3.0	3.3	2.9	4.0	1.9	3.0	3.5	4.3	3.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.0	89.6	89.6	88.0	86.0	84.4	82.2	82.3	84.1	85.7	87.5	88.1
		MCWB	76.8	77.6	78.1	77.8	76.1	75.1	73.2	72.9	74.0	75.4	76.5	76.4
	2%	DB	86.2	87.7	87.7	86.2	84.0	82.1	79.2	80.4	82.3	84.0	85.6	86.2
		MCWB	76.4	77.0	77.3	76.9	75.2	74.0	71.7	72.2	73.5	74.9	75.6	76.0
	5%	DB	85.0	86.0	86.1	84.6	82.4	80.5	78.8	78.9	80.7	82.4	84.0	84.6
		MCWB	76.1	76.5	76.7	76.3	74.6	73.1	71.5	71.1	72.7	74.1	75.3	75.5
	10%	DB	84.2	84.4	84.6	83.9	80.8	79.0	77.3	77.4	79.1	80.8	82.4	84.0
		MCWB	75.8	76.1	76.2	76.0	73.8	72.6	70.7	70.4	71.7	73.4	74.6	75.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.6	79.1	79.2	79.1	77.6	76.4	74.5	74.2	75.5	77.3	78.0	78.6
		MCDB	84.6	85.2	85.2	84.9	83.2	81.4	79.8	80.1	81.8	83.1	84.3	84.2
	2%	WB	77.9	78.2	78.5	78.2	76.4	75.1	73.0	73.0	74.4	76.1	76.7	77.6
		MCDB	83.7	84.2	84.3	83.7	81.6	79.8	77.8	78.1	80.2	81.3	82.6	83.3
	5%	WB	77.2	77.6	77.8	77.5	75.6	74.1	72.3	72.1	73.5	75.2	76.1	76.8
		MCDB	82.8	83.5	83.5	82.8	80.6	78.8	77.1	77.1	78.8	79.9	81.5	82.4
	10%	WB	76.4	76.9	77.1	76.7	74.7	73.2	71.6	71.2	72.6	74.4	75.4	76.2
		MCDB	81.9	82.7	82.7	81.8	79.5	77.8	76.3	76.2	77.8	79.1	80.3	81.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.2	12.2	11.9	11.7	11.4	11.8	11.9	11.9	10.9	10.4	11.2
		MCDBR	12.7	13.2	13.4	13.1	13.0	12.7	12.7	13.1	13.2	12.6	12.8	12.6
	5% WB	MCWBR	5.1	5.2	5.3	5.6	6.1	6.1	6.5	6.5	6.0	5.5	5.0	4.9
		MCWBR	11.9	12.1	12.4	12.2	11.9	11.4	11.9	12.0	12.1	11.4	11.3	11.7
(40) Clear-Sky Solar Irradiance	taub	taub	0.392	0.387	0.386	0.382	0.372	0.361	0.356	0.372	0.423	0.426	0.419	0.411
		taud	2.502	2.521	2.544	2.549	2.563	2.583	2.580	2.502	2.354	2.375	2.422	2.444
	Ebn at Noon	Ebn at Noon	302	301	294	284	276	274	279	283	279	286	292	296
		Edh at Noon	36	36	34	31	29	28	28	32	40	41	39	39
(44) All-Sky Solar Radiation	RadAvg	2085	2040	1842	1607	1362	1269	1341	1534	1690	1817	1794	1976	
	RadStd	162	153	147	124	73	66	63	84	140	202	187	181	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page